# Finite Element Method

## Convergence of the FEM
MECH4103 — Spring 2026

Dr. Behrouz Arash
Associate Professor in Mechanical Engineering

Oslo Metropolitan University

March 4, 2026

- In the assessment of solution quality for various types of elements, a better measure of element performance is needed than the residual for 'eyeballing' the difference between an exact solution and the finite element solution.
- Here, we describe some general methods for quantifying the error in a finite element solution.
- For these purposes, an exact solution is needed, but as we will see in the next lectures, such exact solutions can usually be constructed by 'manufacturing' the solution.
- How can the error in a finite element solution $u^h(x)$ be quantified if we know the exact solution?
- Obviously, comparing the FE solution to the exact solution at a single point may not be helpful. The answer to our question is provided by *norms of functions*.

Using this definition of a norm, we can define the error in a finite element solution by

$$\| e \|_{L_2} = \| u^{\mathrm{ex}}(x) - u^h(x) \| = \left( \int_{x_1}^{x_2} \left( u^{\mathrm{ex}}(x) - u^h(x) \right)^2 \mathrm{d}x \right)^{\frac{1}{2}} , \tag{1}$$

where $u^{\mathrm{ex}}(x)$ is the exact solution and $u^h(x)$ is the finite element solution, so the pointwise error is $u^{\mathrm{ex}}(x) - u^h(x)$. If we think of norms as measures of distance between two functions, then the above is a measure of the distance between the exact and the FE displacement solution.

- The error at any point in the interval contributes to this measure of error because the integrand is the square of the error at any point.
- The above can be considered a *root-mean-square* measure of the error. Thus, the above provides a measure of error that is not affected by a serendipitous absence of error at a few points.
- In comparing errors of different solutions, it is preferable to normalize the error by the norm of the exact solution. The normalized error is given by

$$\bar{e}_{L_2} = \frac{\| u^{\mathrm{ex}}(x) - u^h(x) \|_{L_2}}{\| u^{\mathrm{ex}}(x) \|_{L_2}} = \frac{\left( \int_{x_1}^{x_2} \left( u^{\mathrm{ex}}(x) - u^h(x) \right)^2 \mathrm{d}x \right)^{\frac{1}{2}}}{\left( \int_{x_1}^{x_2} \left( u^{\mathrm{ex}}(x) \right)^2 \mathrm{d}x \right)^{\frac{1}{2}}}. \tag{2}$$

The normalized error can be interpreted quite easily: if the normalized error $\bar{e}_{L_2}$ is of the order of $0.02$, then the average error in the displacement is of the order of $2\%$.

- Although the $L_2$ error in the displacement is quite useful, often we are more interested in the error in the *derivative* of the function.
- For example, in stress analysis, error in the stress, which is proportional to error in the strain, is often of interest.
- In heat conduction, we are often interested in the heat flux.
- Therefore, a more frequently used approach is to compute the *error in energy*:

$$\|\, e \,\|_{\mathrm{en}} = \|\, u^{\mathrm{ex}}(x) - u^{h}(x) \,\|_{\mathrm{en}} = \left( \frac{1}{2} \int_{x_1}^{x_2} E\big(\varepsilon^{\mathrm{ex}}(x) - \varepsilon^{h}(x)\big)^2 \,\mathrm{d}x \right)^{\frac{1}{2}} . \tag{3}$$

- Again, it is preferable in applications to examine the normalized error in energy:

$$\bar{e}_{\mathrm{en}} = \frac{\|\, u^{\mathrm{ex}}(x) - u^{h}(x) \,\|_{\mathrm{en}}}{\|\, u^{\mathrm{ex}}(x) \,\|_{\mathrm{en}}} = \frac{\left( \frac{1}{2} \int_{x_1}^{x_2} E\big(\varepsilon^{\mathrm{ex}}(x) - \varepsilon^{h}(x)\big)^2 \,\mathrm{d}x \right)^{\frac{1}{2}}}{\left( \frac{1}{2} \int_{x_1}^{x_2} E\big(\varepsilon^{\mathrm{ex}}(x)\big)^2 \,\mathrm{d}x \right)^{\frac{1}{2}}} . \tag{4}$$

- When the exact solution is known, the norm of the error in displacements and the energy error are computed easily.
- The integrals are computed by subdividing the domain into elements and then using *Gauss quadrature* in each element.
- In the next example, we will examine the errors as measured by these norms for two elements.
- For this purpose, we will need exact solutions. In one-dimension, exact solutions can easily be obtained for the stress analysis and heat conduction equations.
- In multidimensions, obtaining exact solutions is more difficult, and we will learn how to *manufacture solutions* in next lectures.

We consider a bar of length $2l$, cross-sectional area $A$, and Young's modulus $E$. The bar is fixed at $x = 0$, subjected to linear body force $cx$ and applied traction $\bar{t} = -cl^2/A$ at $x = 2l$ as shown in the figure. The strong form is:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(AE\frac{\mathrm{d}u}{\mathrm{d}x}\right) + cx = 0,$$

$$u(0) = 0,$$

$$\bar{t} = E\frac{\mathrm{d}u}{\mathrm{d}x}\, n\bigg|_{x=2l} = -\frac{cl^2}{A}.$$

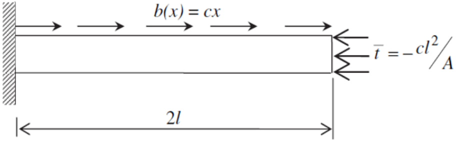



**Figure 5.13** A bar under compression.

A bar under compression.

The exact solution for the above problem can be obtained in closed form:

$$u^{\text{ex}}(x) = \frac{c}{AE}\left(-\frac{x^3}{6} + l^2 x\right), \qquad \varepsilon^{\text{ex}}(x) = \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{c}{AE}\left(-\frac{x^2}{2} + l^2\right).$$

The above problem is solved using the FEM. We study the rate of convergence with **linear** and **quadratic** finite element meshes. The material parameters considered are $E = 10^4\,\mathrm{N\,m^{-2}}$, $A = 1\,\mathrm{m}^2$, $c = 1\,\mathrm{N\,m^{-2}}$, and $l = 1\,\mathrm{m}$.

## Problem Description

We consider a bar of length $2l$, cross-sectional area $A$, and Young's modulus $E$:

- Fixed at $x = 0$ (essential boundary condition: $u(0) = 0$).
- Subjected to a linear body force $cx$.
- Subjected to a traction boundary condition at $x = 2l$:

$$\bar{t} = E\frac{\mathrm{d}u}{\mathrm{d}x}n = -\frac{cl^2}{A}.$$

### 1. Weak Form of the Problem

The strong form of the equilibrium equation is:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(EA\frac{\mathrm{d}u}{\mathrm{d}x}\right) + cx = 0,$$

subject to the boundary conditions:

$$u(0) = 0 \quad \text{(essential)}, \qquad E\frac{\mathrm{d}u}{\mathrm{d}x}\bigg|_{x=2l} = -\frac{cl^2}{A} \quad \text{(natural)}.$$

Multiplying by a weight function $w \in U_0$, integrating by parts, and incorporating the natural boundary condition, the weak form becomes:

$$\int_0^{2l}\left(EA\frac{\mathrm{d}u}{\mathrm{d}x}\frac{\mathrm{d}w}{\mathrm{d}x} - cxw\right)\mathrm{d}x - \bar{t}\,w(2l) = 0.$$

## 2. Discretization and Shape Functions

We discretize the bar into $n_{\mathrm{el}}$ finite elements. Within each element, the displacement $u(x)$ is approximated as:

$$u^e(x) = \mathbf{N}^e \mathbf{d}^e,$$

where $\mathbf{N}^e$ is the shape function matrix (linear or quadratic) and $\mathbf{d}^e$ is the vector of nodal displacements for the element. The derivative is:

$$\frac{\mathrm{d}u^e}{\mathrm{d}x} = \mathbf{B}^e \mathbf{d}^e,$$

where $\mathbf{B}^e$ is the derivative of the shape function matrix.

## 3. Element Matrices

For an element $e$, the stiffness matrix $\mathbf{K}^e$ and force vector $\mathbf{f}^e$ are given by:

$$\mathbf{K}^e = \int_{\Omega^e} (\mathbf{B}^e)^{\mathrm{T}} EA \mathbf{B}^e \, \mathrm{d}x, \qquad \mathbf{f}^e = \int_{\Omega^e} (\mathbf{N}^e)^{\mathrm{T}} cx \, \mathrm{d}x + (\mathbf{N}^e)^{\mathrm{T}} \bar{t} \big|_{x=2l}.$$

### 4. Global System Assembly

The global stiffness matrix $\mathbf{K}$ and global force vector $\mathbf{f}$ are assembled as:

$$\mathbf{K} = \sum_{e=1}^{n_{\text{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{K}^e \mathbf{L}^e, \qquad \mathbf{f} = \sum_{e=1}^{n_{\text{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{f}^e,$$

where $\mathbf{L}^e$ is the transformation matrix mapping local to global degrees of freedom.

### 5. Applying Boundary Conditions

*Essential Boundary Conditions* The displacement at $x = 0$ is fixed: $u(0) = 0$. Partitioning into free $(F)$ and essential $(E)$ DOFs:

$$\begin{bmatrix} \mathbf{K}_{FF} & \mathbf{K}_{FE} \\ \mathbf{K}_{EF} & \mathbf{K}_{EE} \end{bmatrix} \begin{bmatrix} \mathbf{d}_F \\ \mathbf{d}_E \end{bmatrix} = \begin{bmatrix} \mathbf{f}_F \\ \mathbf{f}_E \end{bmatrix} \quad \Rightarrow \quad \mathbf{K}_{FF}\,\mathbf{d}_F = \mathbf{f}_F - \mathbf{K}_{FE}\,\mathbf{d}_E.$$

*Natural Boundary Conditions* The traction at $x = 2l$ contributes to the global force vector:

$$\mathbf{f}_{\text{traction}} = (\mathbf{N}^e)^{\mathrm{T}} \left( -\frac{cl^2}{A} \right).$$

### 6. Reaction Forces

After solving for the displacements $\mathbf{d}$, the reaction forces are:

$$\mathbf{R} = \mathbf{K}\mathbf{d} - \mathbf{f}.$$

*Linear Elements*

$$\mathbf{N}^e = \begin{bmatrix} N_1^e & N_2^e \end{bmatrix} = \frac{1}{l^e} \begin{bmatrix} x_2^e - x & x - x_1^e \end{bmatrix}, \qquad \mathbf{B}^e = \begin{bmatrix} \frac{dN_1^e}{dx} & \frac{dN_2^e}{dx} \end{bmatrix} = \frac{1}{l^e} \begin{bmatrix} -1 & +1 \end{bmatrix}.$$

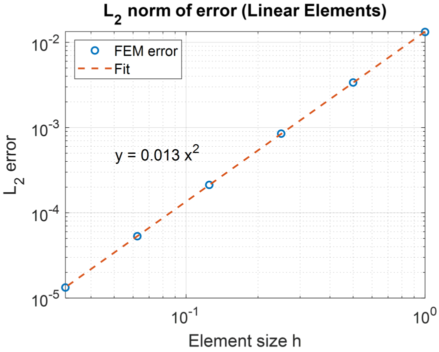
*Quadratic Elements*

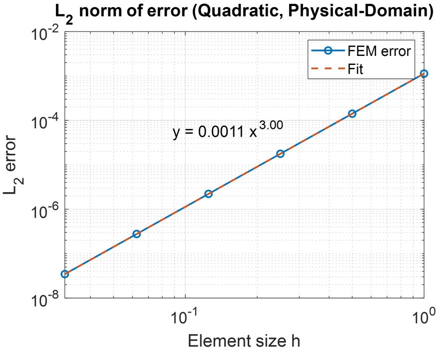
$$\mathbf{N}^e = \frac{2}{(l^e)^2} \begin{bmatrix} (x - x_2^e)(x - x_3^e) & -2(x - x_1^e)(x - x_3^e) & (x - x_1^e)(x - x_2^e) \end{bmatrix},$$

$$\mathbf{B}^e = \frac{2}{l^e} \begin{bmatrix} 2x - x_2^e - x_3^e & -2(2x - x_1^e - x_3^e) & 2x - x_1^e - x_2^e \end{bmatrix}.$$

# Convergence by Numerical Experiments

- The figures below show the log of the error norm as a function of the log of element size $h$.
- As can be seen from these results, the log of the error varies *linearly* with element size and the slope depends on the order of the element and whether the error is in the function or its derivative.



- If we denote the slope by $\alpha$, then the error in the function (the $L_2$ norm) can be expressed as

$$\log(\| e \|_{L_2}) = C + \alpha \log h, \tag{5}$$

where $C$ is an arbitrary constant (the $y$-intercept) and $\alpha$ is the *rate of convergence*. Taking the power of both sides:

$$\| e \|_{L_2} = Ch^{\alpha}. \tag{6}$$

- For linear two-node elements, $\alpha = 2$, whereas for quadratic elements, $\alpha = 3$. The error for the two-node element is *quadratic*; the three-node element is of *third order*.
- The constant $C$ depends on the problem and the mesh, and it is not of much importance.
- The crucial concept to be learned from this equation is how the error decreases with element size.
- From (6): if the element size is halved, the error in the function decreases by a factor of $4$ and $9$ for linear and quadratic elements, respectively.
- The formula has been generalized: if a finite element contains the complete polynomial of order $p$, the error in the $L_2$ norm varies according to

$$\| e \|_{L_2} = Ch^{p+1}. \tag{7}$$

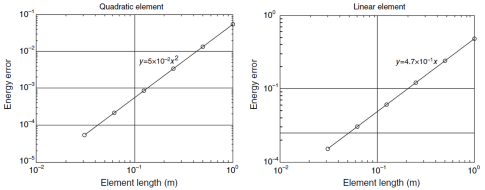
- The slope of the convergence plot for derivatives, i.e. the error in energy, is *one order lower*. For an element complete up to order $p$:

$$\| e \|_{\mathrm{en}} = Ch^p. \tag{8}$$

- Thus, the accuracy in the derivative is one order less than the accuracy in the function. If the element size is halved, the error in energy decreases by factors of $2$ and $4$ for linear and quadratic elements, respectively.

**Key takeaway**

Quadratic elements give more accuracy for the buck. In linear analysis, quadratic elements are almost always preferred — their advantages in accuracy are overwhelming and come at little cost.




Energy norm of error for linear (left) and quadratic (right) finite element meshes.

Energy norm of error for quadratic (left) and linear (right) FE meshes.

**Next: Isoparametric Formulation**

**Thanks for your attention!**